

Fresh deep critical audit of Parts I and II

Proof-breaking issues, repairs, and source-scope verification

Independent reconstruction pass

16 August 2026

Bottom line.

Part I. The principal implication chain

$$\begin{aligned} \text{GJS CE1/CE2} &\implies M_2 \implies \text{uniform cutoff gap} \implies \text{LPPL} \\ &\implies \text{full local-norm net limit} \implies \text{spectral ground state.} \end{aligned}$$

is mathematically coherent. I found one omitted localization decomposition in the CE2-to-gap step and one missing mollifier hypothesis in Appendix C. Both have short, fully explicit repairs. The remaining substantive caveat is source scope: GJS Theorem 1.1.8 is printed as uniform “as $h \uparrow 1$ ”, while Part I quantifies over its entire enlarged cutoff class. GJS p. 629 says the named estimates are uniform in the space cutoff, which strongly supports the intended use, but the manuscript’s Appendix C does not reproduce enough of the cluster expansion to constitute an independent proof of every strengthened quantifier. The Main Theorem is therefore **CONDITIONAL** on that exact uniformity reading unless its cutoff class is narrowed.

Part II. The current final theorem is **FAIL** as a proof in the files as written. The load-bearing shifted-interaction comparison (V2, Proposition 6.5) uses an ambiguous cross power symbol, introduces an undefined R_{tail} , and then drops it. This breaks A6 and hence gate (II.11). Separately, V2 Proposition 1.2 contains a false combinatorial assertion: four edge momenta can conserve momentum. That second error does *not* change the advertised rate; its missing contribution is explicitly bounded below. The companion Lamport reconstruction supplies the missing Fourier partition and a complete repair of Proposition 6.5, so the theorem is salvageable, but the repair is new mathematics and must be adopted before Part II can be called proved.

Contents

1	Scope, freshness, and standards	2
1.1	Documents of record	2
1.2	Validity standard	2
1.3	Reproducibility artifacts	3
2	Severity-ranked findings	3
2.1	F-II-02: the shifted-interaction comparison is not proved	3
2.1.1	Why the displayed decomposition is not an identity	3
2.1.2	A complete repair	3
2.2	F-II-01: the “three or more edge momenta are impossible” claim is false	5

2.3	F-I-02: CE2 is applied to a nonlocalized polynomial	5
2.4	F-II-03: V3's periodic N2 rate is asserted, not derived	6
2.5	F-I-01: GJS cutoff uniformity is stronger than the theorem's literal wording	7
2.6	F-I-03: the ultraviolet mollifier lacks used hypotheses	7
2.7	F-II-04: a constant has unstated anisotropic-grid dependence	7
2.8	F-II-05: finite-mode transplant is too compressed for the stated standard	7
3	Part I proof-chain audit	8
3.1	Imports and dimensionless reduction	8
3.2	Slab limit and uniform moment	8
3.3	Uniform gap	8
3.4	Path regularity, filter, and LPPL	8
3.5	Full net, normality, and spectral passage	8
3.6	Dichotomy scope	9
4	Part II proof-chain audit	9
4.1	Correct target and free obstruction	9
4.2	V1: thermal representations	9
4.3	V2: unshifted interaction and Wick dictionary	9
4.4	V3: continuum Hamiltonian package	9
4.5	V4: assembly and quantifiers	10
5	External-source verdicts	10
6	Final disposition	11

1 Scope, freshness, and standards

1.1 Documents of record

The Part I document of record is `part_i/main.tex`. The current Part II proof chain is the model sheet, free fixed-coarse rate, quadratic obstruction, projective compactness, Weyl reduction, gate scoping note, and V1–V4. The older M0–M6 files and earlier audits were used only to locate possible source pages; no earlier verdict was inherited. Every cited page was rendered again from the local PDF.

1.2 Validity standard

A step passes only if its domains, quantifiers, normalizations, topology, and limiting order are specified and every displayed equality follows from stated premises. A true result with a dropped term is still a proof defect. A source citation passes only if the exact cited source page states the needed result at the needed scope. Numerical checks are permitted only to falsify an identity, never to establish a theorem.

1.3 Reproducibility artifacts

The claim, dependency, source, citation, and finding ledgers are CSV files in `ledgers/`. The source-evidence companion contains 88 freshly rendered pages. The Lamport reconstruction gives the repaired proof chain with hierarchical steps.

2 Severity-ranked findings

2.1 F-II-02: the shifted-interaction comparison is not proved

Audit finding 2.1 (Dropped Fourier sectors in V2 Proposition 6.5). **Severity:** **CRITICAL**.

V2 lines 1405–1417 assert a decomposition of the lattice, mixed, and continuum quadratic forms, add a term called R_{tail} , describe it only as “regrouping” missing modes, and then omit it from the inequality at lines 1419–1422. In addition, Lemma 6.4 defines the cross power symbol on the j -fold symmetric band but does not put the edge weights χ into the defining convolution. Its proof treats the lattice symbol but does not separately prove the symmetric-edge cross case.

The omission is load-bearing: Proposition 6.5 is used by V4 Lemma 5.2, Theorem A6, gate (II.11), and the final fixed-coarse theorem.

2.1.1 Why the displayed decomposition is not an identity

Put $R = \pi/\varepsilon_N$ and let

$$B_N = \{-R, -R + \pi/L, \dots, R - \pi/L\}, \quad [k]_N \in B_N, \quad k - [k]_N \in 2R\mathbb{Z}.$$

Write $b_N(p)$ for the spatial Fourier coefficient of a lattice coefficient g_N , $p \in B_N$, and $b_\infty(k)$ for the continuum coefficient. If $K_{\times,j}(\kappa)$ is the exact-output j -fold symbol of the translation-invariant surrogate mixed kernel, direct Fourier expansion gives

$$\langle g_N, C_\times^j g_\infty \rangle = \frac{1}{2Lt} \sum_{\kappa} K_{\times,j}(\kappa) \overline{\tilde{b}_N(-[k]_N, -k_0)} \tilde{b}_\infty(k, k_0). \quad (2.1)$$

The sum runs over the entire j -fold output band, not only $k \in B_N$. Thus outputs $k = p + 2Rn$ with $n \neq 0$ pair the lattice coefficient $b_N(-p)$ with the continuum coefficient $b_\infty(p + 2Rn)$. They do not occur in the band-limited quadratic form used at V2 line 1408. Those are precisely mixed alias-output terms; calling them R_{tail} does not make them vanish.

There is a second, independent correction. The true mixed covariance is $C_\times + R_N$ and is not translation invariant in (x, y) . It may be bounded in physical space, as V2 lines 1362–1388 do, but it cannot simultaneously be represented by the unweighted cross symbol of Lemma 6.4. One must choose one of the two descriptions and account for the difference exactly once.

2.1.2 A complete repair

The following partition closes the gap. It is proved step by step in the Lamport reconstruction; the estimates are recorded here so the audit verdict is falsifiable.

Define, for $p \in (2\pi/t)\mathbb{Z} \times B_N$,

$$\begin{aligned} K_{N,j}^{\text{fold}}(p) &= (2Lt)^{-(j-1)} \sum_{\substack{\kappa_i \in D_N \\ \sum k_{0,i}=p_0, [\sum k_i]_N=p}} \prod_i \hat{c}_N(\kappa_i), \\ K_{\times,j}(\kappa) &= (2Lt)^{-(j-1)} \sum_{\substack{\kappa_i \in D_N \\ \sum \kappa_i=\kappa}} \prod_i \hat{c}_{\times}(\kappa_i), \\ K_{\infty,j}(\kappa) &= (2Lt)^{-(j-1)} \sum_{\sum \kappa_i=\kappa} \prod_i \hat{c}_{\infty}(\kappa_i). \end{aligned}$$

All sums are nonnegative. The envelope-convolution lemma implies, for $j = 1, 2, 3$,

$$\sup_p |K_{N,j}^{\text{fold}}(p) - K_{\infty,j}(p)| \leq C\varepsilon_N^2 \log_N^2, \quad (2.2)$$

$$\sup_{p \in B_N} |K_{\times,j}(p) - K_{\infty,j}(p)| \leq C\varepsilon_N^2 \log_N^2, \quad (2.3)$$

$$\sup_{|k| \geq R} K_{\times,j}(k_0, k) \leq C\varepsilon_N^2 \log_N^{j-1}. \quad (2.4)$$

For (2.2), partition the folded lattice configurations into exact output and the finitely many shifts $2Rn$. Telescope the exact-output products one factor at a time, use

$$|\hat{c}_N - \hat{c}_{\infty}|(u) \leq C\varepsilon_N^2 \frac{u_{\text{sp}}^4}{(m^2 + |u|^2)^2},$$

and sum the remaining $j - 1$ envelope factors. Every nonzero shift has one convolution argument at distance at least R and hence contributes (2.4). Continuum configurations with an input outside B_N are bounded by fixing that input and applying the same tail convolution. This proves (2.2). The proof of (2.3) is identical without folded output. Equation (2.4) is the convolution-decay estimate at distance R .

Extend b_N by zero outside the fundamental band and put $\delta b = b_N - b_{\infty}$ on B_N , $\delta b = -b_{\infty}$ outside. Splitting (2.1) into $n = 0$ and $n \neq 0$ gives the exact identity

$$\begin{aligned} Q_{N,j} - 2Q_{\times,j} + Q_{\infty,j} &= \langle \delta b, K_{\infty,j} \delta b \rangle + \langle b_N, (K_{N,j}^{\text{fold}} - K_{\infty,j}) b_N \rangle \\ &\quad - 2\langle b_N, (K_{\times,j} - K_{\infty,j}) b_{\infty,B} \rangle - 2A_{\times,j}, \end{aligned} \quad (2.5)$$

where

$$A_{\times,j} = \frac{1}{2Lt} \sum_{p_0 \in (2\pi/t)\mathbb{Z}, p \in B_N, n \neq 0} K_{\times,j}(p_0, p + 2Rn) \overline{b_N(-p, -p_0)} b_{\infty}(p + 2Rn, p_0).$$

Only finitely many n occur because $K_{\times,j}$ is supported in the j -fold band. Cauchy–Schwarz and (2.4) give

$$|A_{\times,j}| \leq C\varepsilon_N^2 \log_N^2 \|b_N\|_2 \|b_{\infty}\|_2. \quad (2.6)$$

The physical-space edge remainder satisfies

$$|Q_{\text{true cross}} - Q_{\times,j}| \leq C(2Lt)^2 \Sigma_M(p)^{2(4-j)} \varepsilon_N \log_N^{j-1}. \quad (2.7)$$

Because $\varepsilon_N = \varepsilon_M 2^{-r}$ and $\theta_H = (1 + \eta)/2 < 1$,

$$\varepsilon_N \log_N^2 \leq C(\varepsilon_M, \eta) 2^{-r\theta_H}. \quad (2.8)$$

Finally, the power coefficients obey

$$\|b_N^{(h_N^{4-j})} - b_{\infty}^{(h_{\infty}^{4-j})}\|_2^2 \leq C 2^{-r\theta_H}. \quad (2.9)$$

To prove (2.9), telescope the $(4 - j)$ -fold convolution; apply $\ell^2 * \ell^1 \rightarrow \ell^2$ Young inequality to the ordinary differences; for lattice aliases, $|k_1 + \dots + k_{4-j}| \geq R$ forces one $|k_i| \geq R/(4 - j)$, so put that factor in the $D4 \ell^2$ tail and the remaining factors in ℓ^1 . The same argument treats missing continuum modes.

Equations (2.2)–(2.9), inserted into the Wick–Isserlis identity, yield

$$\|\Delta\mathcal{U}_N - \Delta\mathcal{U}\|_2 \leq C\{2^{-r(1+\eta)/4} + \varepsilon_N \log_N\},$$

which is exactly the claimed rate. Thus the proposition is repairable, but the repair is absent from V2 and cannot be inferred from its undefined R_{tail} .

2.2 F-II-01: the “three or more edge momenta are impossible” claim is false

Audit finding 2.2 (Omitted four-edge sector). **Severity: MAJOR.**

V2 Proposition 1.2(iii)(b) says that a fourth-moment cross configuration with at least three edge momenta is impossible. Let $R = \pi/\varepsilon_N$. The symmetric band contains both $+R$ and $-R$, and

$$(+R) + (+R) + (-R) + (-R) = 0.$$

Hence six ordered four-edge sign configurations exist. The exact cross-moment formula at V2 lines 577–579 is correct; the subsequent classification is not exhaustive.

Put $f(k_0) = \hat{c}_\times(k_0, R)$; evenness makes the sign irrelevant. Each edge has weight $2^{-1/2}$, so the omitted positive contribution to $\mathbb{E}[X_4 Y_4]$ is

$$\Delta_{4e} = 4!(2Lt)^{-2} \frac{3}{2} \sum_{k_{0,1}+\dots+k_{0,4}=0} \prod_{i=1}^4 f(k_{0,i}). \quad (2.10)$$

The factor $3/2$ is $\binom{4}{2}(2^{-1/2})^4 = 6/4$.

Let $A = \sum f$ and $B = \sum f^2$. Since $f(k_0) \leq (k_0^2 + \gamma_e^2)^{-1}$, $\gamma_e \geq 2\varepsilon_N^{-1}$, and $t\gamma_e/2 \geq 2t_0$,

$$A \leq \frac{t}{2\gamma_e} \coth(t\gamma_e/2) \leq \frac{t\varepsilon_N}{4} \coth(2t_0),$$

$$B \leq \gamma_e^{-2} A \leq \frac{t\varepsilon_N^3}{16} \coth(2t_0).$$

Furthermore

$$\sum_{k_{0,i}=0} \prod_i f(k_{0,i}) = \|f * f\|_2^2 \leq \|f * f\|_\infty \|f * f\|_1 \leq BA^2.$$

Substitution into (2.10) gives the explicit bound

$$0 \leq \Delta_{4e} \leq \frac{9t}{256L^2} \coth(2t_0)^3 \varepsilon_N^5. \quad (2.11)$$

This is smaller than the $O(t\varepsilon_N^3 \log_N^2)$ correction already allowed. Because it increases the cross moment, dropping it also decreases the true squared difference relative to the approximate expression. Thus (2.11) repairs the equality and preserves the rate; it does not by itself destroy A4(a).

2.3 F-I-02: CE2 is applied to a nonlocalized polynomial

Audit finding 2.3 (Missing source-required cell decomposition). **Severity: MAJOR** in the written proof; short repair.

Part I Lemma `ce2strip` applies GJS Theorem 1.1.7 directly to Q . The source theorem is stated for two localized monomials. GJS p. 594 explicitly says that a nonlocalized/general polynomial is written as a sum of localized monomials and defines its norm as the sum of their norms. Part I established this norm but omitted the decomposition at the point of use.

Write the finite source-prescribed decomposition $Q = \sum_{a \in A} Q_a$. Translation to either time row does not change the component norms. Every localization square of $Q_a(-t/2)$ is separated from every square of $Q_b(t/2)$ by width at least $t - 2$. Therefore CE2 gives

$$\begin{aligned} & |\text{Cov}_{dq_h}(Q(-t/2), Q(t/2))| \\ & \leq \sum_{a,b} |\text{Cov}_{dq_h}(Q_a(-t/2), Q_b(t/2))| \\ & \leq C_7 e^{-m_1(t-2)} \sum_{a,b} \|Q_a\|_{\text{GJS}} \|Q_b\|_{\text{GJS}} \\ & = C_7 e^{2m_1} e^{-m_1 t} \left(\sum_a \|Q_a\|_{\text{GJS}} \right)^2 = C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}}^2. \end{aligned}$$

This is exactly the displayed Lemma 7.4 estimate. After insertion, semigroup decay and the uniform-gap theorem are restored without changing a constant.

2.4 F-II-03: V3's periodic N2 rate is asserted, not derived

Audit finding 2.4 (Unwritten two-dimensional tail estimate). **Severity: MAJOR in a no-compression proof; repairable.**

V3 Proposition 2.6(iii) states that the periodic spatial truncations satisfy the N2 rate by “the same envelope tail counts,” without writing the k_0 sum or the constrained four-fold convolution. V3 Lemma 6.1 then consumes that rate in the continuum marked-slice limit.

For the quartic chaos, Wick–Isserlis and a union bound over the first out-of-band input give

$$\begin{aligned} \|Y_4^{(K')} - Y_4^{(K)}\|_2^2 & \leq C(2Lt)^{-2} \sum_{|k| > K, k_0} \hat{c}(k_0, k) F_3(-k_0, -k) \\ & \leq C \sum_{|k| > K, k_0} \frac{(1 + \log(2 + |k_0| + |k|))^2}{(m^2 + k_0^2 + k^2)^2}. \end{aligned}$$

For $t \in [t_0, T]$, the one-dimensional grid comparison gives

$$\sum_{k_0 \in (2\pi/t)\mathbb{Z}} \frac{(1 + \log(2 + |k_0| + |k|))^2}{(m^2 + k_0^2 + k^2)^2} \leq Ct \frac{(1 + \log(2 + |k|))^2}{(m^2 + k^2)^{3/2}}.$$

The spatial tail is bounded by

$$\sum_{|k| > K} \frac{(1 + \log(2 + |k|))^2}{(m^2 + k^2)^{3/2}} \leq CL \frac{(1 + \log(1 + K/m))^2}{K^2}.$$

Thus the quartic norm is $O(\sqrt{tL} K^{-1} \log K)$, stronger than V3's $O(K^{-1/2} \log^2 K)$. The quadratic chaos uses $\sum_{|k| > K, k_0} \hat{c}(k_0, k)^2$ and has the same or better rate. The difference between the truncated thermal-to-vacuum Wick constants is exponentially small because

$$0 \leq \frac{\coth(t\gamma/2) - 1}{2\gamma} = \frac{1}{\gamma(e^{t\gamma} - 1)} \leq C(t_0, m) e^{-t_0\gamma} / \gamma.$$

The claimed N2 estimate follows. This derivation is included in the reconstruction.

2.5 F-I-01: GJS cutoff uniformity is stronger than the theorem’s literal wording

Audit finding 2.5 (Source-quantifier ambiguity). **Severity: MAJOR** if full-class uniformity is not accepted; otherwise citation qualification.

GJS Theorem 1.1.7 is explicitly independent of h . Theorem 1.1.8 is printed as a bound holding uniformly “as $h \uparrow 1$.” Part I promotes it to every $h \in \mathcal{C}_E$, including arbitrary compact slabs and affine cutoff paths. GJS p. 629 states that the estimates of Theorems 1.1.8, 1.1.11, 3.1 and 4.1 are uniform in the space cutoff h . This is strong positive support, but it is not a verbatim restatement of the enlarged quantifier used in Part I.

Appendix C’s “structural uniformity” proposition lists the locations where h enters the cluster expansion, but it does not reproduce the polymer estimates, convergence summations, or constants. It is therefore a source audit, not a standalone proof of a strengthened cluster-expansion theorem. There are two rigorous ways to close the issue:

1. state as an explicit imported assumption that GJS p. 629’s “uniform in the space cutoff h ” covers the entire source cutoff class used here; or
2. narrow the manuscript cutoff family to a directed family covered literally by the printed limiting statement and prove that every slab/path application belongs to a uniform tail of that family.

The present manuscript does neither with complete formal precision. This is why the Part I verdict is conditional rather than an unconditional fail: the source evidence is substantial and the internal argument needs only the bound, but the exact import scope must be made explicit.

2.6 F-I-03: the ultraviolet mollifier lacks used hypotheses

Audit finding 2.6 (Approximate identity underspecified). **Severity: MODERATE.**

Part I Appendix C fixes a “smooth approximate identity” δ_κ and later treats $G_\kappa^{(1)}$ and G_κ as averages of G over balls of radius at most $2/\kappa$. This requires $\delta \geq 0$, $\int \delta = 1$, compact support, the scaling $\delta_\kappa(x) = \kappa^2 \delta(\kappa x)$, and for the symmetric Wick-covariance identities it is convenient to require $\delta(x) = \delta(-x)$. None is stated.

Choose once and for all $\delta \in C_c^\infty(B(0, 1))$, $\delta \geq 0$, $\delta(x) = \delta(-x)$, $\int \delta = 1$, and define $\delta_\kappa(x) = \kappa^2 \delta(\kappa x)$. Then the “average over balls” bounds and all covariance identities used in the proof follow. This repairs the lemma. It does not independently repair the broader cluster-expansion uniformity issue F-I-01.

2.7 F-II-04: a constant has unstated anisotropic-grid dependence

V2 Lemma 3.1 describes certain convolution constants as depending only on m . The proof absorbs isolated terms by $tL \geq 1$ and uses a lower bound $t \geq t_0$; hence the constant also depends on the grid control, equivalently on t_0 (and on fixed L if the condition is not normalized). Downstream statements already allow that dependence, so this is not proof-breaking. Correct the displayed dependency.

2.8 F-II-05: finite-mode transplant is too compressed for the stated standard

V3 Theorem 3.1 transfers the model-sheet and V1 finite-dimensional kernel results into the continuum-mode truncations. The spaces are finite-dimensional, the covariance matrices are positive, and the Mehler/Trotter argument is valid; no contradictory normalization was found. Nevertheless, the theorem repeatedly says that earlier steps are “transplanted” instead of writing the unitary coordinate identification, reference Gaussian density, and trace-kernel normalization. The Lamport reconstruction supplies those steps. This is a rigor/presentation defect, not an identified false result.

3 Part I proof-chain audit

3.1 Imports and dimensionless reduction

The cutoff Hamiltonian existence/uniqueness, FKN formula, stability, hypercontractivity, domain smoothing, finite propagation, and covariance claims have exact positive source pages in the evidence dossier. Three former pinpoint risks are resolved:

1. Simon III.6 and V.14–V.15 are the FKN results; III.12 is not.
2. Simon I.17 and III.17 are the hypercontractive results; V.10 is not the direct source of the displayed abstract inequality.
3. Glimm–Jaffe Expositions Theorem 7.4 is the model-specific domain-smoothing theorem; Theorem 7.3 alone has an extra bounded-below premise.

The dilation in Part I tracks the mass, coupling, spatial cutoff, Wick covariance, and observable norm consistently. No reversed small-coupling inequality was found.

3.2 Slab limit and uniform moment

The bounded FKN formula is extended first to signed bounded functions, then to unbounded polynomials by truncation and uniform square moments. Simplicity alone identifies the large-slab rank-one limit. The M_2 proof uses CE1 on $V(h)^2$, which is permitted as a product of individually Wick-ordered monomials; no Wick recombination is needed. The single overall K_1 in the GJS norm is used correctly.

3.3 Uniform gap

After inserting F-I-02's cell decomposition, the logic is sound. The Euclidean two-time bound gives $\langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq C_Q e^{-m_1 t}$ on a dense family. If $P_{(0,m_1)} \neq 0$, choose a vector in its range and approximate it by that dense family; the spectral theorem gives a lower exponential rate incompatible with the upper rate. Simplicity identifies the zero-energy projection. The proof does not infer a gap from clustering for only one observable; it first proves density, which is the necessary step.

3.4 Path regularity, filter, and LPPL

The affine path stays within the admissible cutoff class. The semigroup derivative is proved in a weak pairing and upgraded using the uniform M_2 control. The spectral projection derivative is licensed by the common uniform gap. The exact filter has the correct Fourier convention and equals $H^{-1}(1 - P)$ on the gapped subspace. Finite propagation localizes the truncated filter, while the time tail is controlled by the filter decay. The unbounded interaction is used only through affiliation and the uniform second moment. Shell counting in one space dimension produces a summable, super-polynomial LPPL rate. No hidden assumption of bounded interaction was found.

3.5 Full net, normality, and spectral passage

The plateau-containment relation is technically a preorder rather than an antisymmetric partial order, but directed-net convergence is unaffected. Local dual spaces are complete, restriction limits are compatible, and positivity extends to the quasi-local completion. Normality is proved from norm convergence to normal cutoff vector states; the monotone-net argument is valid. The ground-state spectral condition is passed using exact equality with a nonnegative cutoff spectral measure on every

fixed compact time window, followed by an L^1 tail limit. This avoids an unjustified convergence of Hamiltonians.

3.6 Dichotomy scope

The two-phase exclusion theorem is explicitly conditional on hypothesis B4. The cited GJS phase papers prove phase coexistence and pure-phase gaps in their stated regimes but do not supply the classification of every parity-invariant Hamiltonian ground state required by B4. Because the manuscript now labels B4 as a hypothesis, this is not a false theorem; it must not be advertised as an unconditional uniqueness result.

4 Part II proof-chain audit

4.1 Correct target and free obstruction

The quadratic obstruction correctly rejects a diagonal finest-scale local-state norm target. MMST's construction is projective: convergence is evaluated after pullback to one fixed coarse algebra. The localized high-frequency sequence exposes the difference between pointwise/fixed-coarse weak-star convergence and a supremum over a moving unit ball. The projective compactness theorem and full-sequence Weyl criterion are standard compactness/density arguments and are correct.

The exact D4 finite product, dispersion comparison, and fixed-coarse free rate were recomputed. The support, normalization, and momentum/position covariance factors agree with MMST. The cited source proves convergence but not the new rate; the rate is supplied internally and its summations are adequate.

4.2 V1: thermal representations

The finite-dimensional Schrödinger realization produces the correct Weyl phase. The Mehler kernel normalization integrates to the harmonic trace. Completing the square in the bridge variables yields the stated harmonic jump interpolant and action. The V4 erratum correctly specifies the periodic Gaussian reference measure, avoiding a circular use of the interacting measure. Mode products converge under the D4 envelope, and all uses of the thermal rate occur for $t \geq t_* > 0$.

4.3 V2: unshifted interaction and Wick dictionary

The same-noise coupling needs the special edge symmetrization because the half-open spatial band is not invariant under $k \mapsto -k$. The marginal lattice covariance is correct. The unshifted A4(a) comparison has valid dispersion, missing-mode, and lattice alias estimates once the four-edge term (2.10) is added. The Wick recombination formula and the vacuum/thermal dictionary constants have the correct signs; no residual mass counterterm remains under the matched convention.

The shifted comparison is the unique unresolved load-bearing point in the original file; see F-II-02.

4.4 V3: continuum Hamiltonian package

The finite-mode FKN semigroups are uniformly lower bounded by Nelson estimates. Strong semigroup convergence defines a self-adjoint generator; the cylinder-core generator identity and lower form bound are compatible. The hypercontractive ladder upgrades a single-time Hilbert–Schmidt factor to trace class and permits trace-norm finite-mode convergence. Eigenvalue and partition-function convergence then follow from standard trace-ideal arguments. The periodic FKN

clause requires the quantitative tail repair in F-II-03. Positivity improving follows from Simon I.16 because $\|e^{-t\gamma}\| = e^{-tm} < 1$; simplicity and a positive first gap follow from compact resolvent/trace class. No use at norm 1 was found.

4.5 V4: assembly and quantifiers

Every invocation of V2's $tL \geq 1$ estimates occurs with $t \geq \max(1, 1/L)$. The corrected Laplace-transfer lemma needs transform convergence only for $t \geq t_*$, and V4 uses exactly that range. The interchange lemma requires a uniform spectral tail, supplied by the transferred gap and normalized partition bound. Thus the order

$$\text{fixed } t : N \rightarrow \infty, \quad \text{then } t \rightarrow \infty$$

is legitimate once A6 is proved. Because A6 currently depends on the incomplete V2 Proposition 6.5, the final theorem is not proved in the original chain. With the repaired Proposition 6.5 and F-II-03 inserted, the remaining assembly closes.

5 External-source verdicts

Source/result	Exact evidence	Positive conclusion	Limitation
GJ I-II	I02–I08	cutoff core, ground state, covariance, locality	no full-net vacuum convergence
GJ Expositions 7.4	I24–I25	model-specific semigroup domain smoothing	7.3 alone has extra premise
GJS 1.1.7	I31	CE2 and explicit h independence	localized monomials; decompose general Q
GJS 1.1.8/p.629	I32–I33	CE1 and space-cutoff uniformity mechanism	full-class quantifier should be stated as import
GRS II.14/II.16	I34–I35	FKN and general marked-time FKN	none relevant
Simon I.17/III.17	I44–I45/I48–I49	abstract and time-region hypercontractivity	V.10 is not the direct pinpoint
Simon V.7/V.14–15	I50/I54–I55	stability and interacting FKN	none relevant
MMST 4.1–4.5	II07–II10	observable embedding and free fixed-coarse limit	no interacting theorem or rate
MMST Section 6.1	II14	identifies interacting extension as future work	cannot be used as theorem
Battle–Federbush	II15–II17	phase-cell wavelet mechanism	different variables; no A4/A5 estimates
NRSS 4.1	II19	anharmonic LR under Fourier/ L^1 assumptions	Wick quartic fails $V' \in L^1$

6 Final disposition

Deliverable theorem	Status	Exact disposition
Part I Main Theorem	CONDITIONAL	Internal chain closes after the explicit CE2 decomposition. Accept GJS p. 629 as a full cutoff-class import or narrow the class. State the mollifier hypotheses in Appendix C.
Part I canonical-vacuum corollary	CONDITIONAL	Follows from the Main Theorem and the verified GJ III construction formulas.
Part I two-phase exclusion	CONDITIONAL	Correct only under explicit B4; cited phase papers do not discharge B4.
Part II unshifted A4(a)	REPAIRABLE GAP	Add the four-edge contribution (2.10)–(2.11). The rate survives.
Part II shifted A4(c)/Proposition 6.5	FAIL as written	Undefined/dropped mixed alias sectors break the proof. Equations (2.1)–(2.9) provide a complete replacement.
Part II A6 and gate (II.11)	FAIL as written	Depend on Proposition 6.5. They become valid after the replacement and the explicit V3 periodic-tail derivation.
Part II final fixed-coarse theorem	FAIL as written; salvageable	The free target, compactness, Weyl reduction, V1, and V3/V4 architecture are sound. Adopt the two repairs before asserting the theorem.

No evidence was found that the cited literature already proves the new interacting fixed-coarse Part II comparison. The value of the present audit is therefore not a source substitution: it isolates the exact internal Fourier estimate on which the new result stands or falls and supplies a checkable repair.